

Improved Noise Bound in BFV Homomorphic Encryption and Its Application to Multiplication

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Abstract. Fully Homomorphic Encryption (FHE) enables computations on encrypted data without requiring decryption. However, each computation increases the noise level, which can eventually cause decryption failures once a certain threshold is reached. In particular, homomorphic multiplication significantly amplifies noise in the ciphertext. In this work, we revisit Ring-learning-With-Error (RLWE) based encryption proposed by Fan et al. and present an optimized noise growth approach by swapping the sample space for secret key and error distribution. Thereafter, we revisit BFV homomorphic multiplication proposed by Kim et al. (ASIACRYPT'21) and present an optimized noise bound. Later, we empirically check the hardness of proposed scheme using lattice estimator. Our analysis demonstrates that the proposed method achieves more than 128-bit security and achieves a lower noise bound than existing techniques.

Keywords: BFV · Fully Homomorphic Encryption (FHE) · Multiplication · Noise Bound · RLWE.

1 Introduction

1.1 Background

The notion of homomorphic encryption allows computations to be performed on encrypted data without requiring the secret key [1,2]. A significant breakthrough in this area was Gentry's 2009 construction of fully homomorphic encryption (FHE), which enables arbitrary computations on ciphertexts while preserving encryption security [3]. The most common FHE schemes are categorized into three types. The first type presents Fastest Homomorphic Encryption in the West FHEW and Torus Fully Homomorphic Encryption (TFHE) that deals with lookup tables and Boolean circuits (also known as the Chillotti, Gama, Georgieva, and Izabachène (CGGI) scheme) [4,5,6,7,8]. The second type presents Brakerski-Gentry-Vaikuntanathan (BGV) and Brakerski-Fan-Vercauteren (BFV) encryption scheme that deals with modular arithmetic over finite fields (enabling single-instruction-multiple-data (SIMD) functionality) [9,10]. Thereafter,

the third type presents Cheon-Kim-Kim-Song (CKKS) scheme that deals with approximated computation of real and complex numbers. The security of all these schemes is based on the Ring Learning With Errors (RLWE) assumption [11], where noise is inherently added during encryption. This noise accumulates as computations are performed on encrypted data, and once it exceeds a certain threshold, further operations result in decryption failure. Thus, to accommodate noise in all types of FHE schemes, the ciphertext modulus (say Q) needs to be large enough, or a special bootstrapping algorithm is designed to reduce the noise growth and keep the value of Q relatively small.

In this work, we focus on BFV scheme.⁴ The BFV scheme is scale-invariant that encodes the message in most significant bits (MSBs) of integers in $\mathbb{Z}_Q$, and noise in the least significant bits (LSBs). However, each arithmetic operation increases the noise level in the ciphertext. Among arithmetic operations, multiplication is more expensive as compared to addition (as noise growth is quadratic in multiplication [9,10]). In multiplication, ciphertext polynomials are multiplied without modular reduction and thereafter scaling with respect to t/Q (where t is plaintext modulus) is performed.

The primary objective of this paper is to mitigate noise growth in BFV ciphertext multiplication. Specifically, in this work we first revisit the Fan et al. [12] scheme and introduce an improvisation by swapping the sample space size for secret key and error distribution. Later, we empirically show its impact on Kim et al. [13] BFV homomorphic multiplication. Our approach provides better noise bound, thereby enhancing the practicality of homomorphic computations by delaying bootstrapping.

1.2 Literature Review

In this Section, we discuss the relevant literature for multiplication of two BFV ciphertexts.

To the best of our knowledge, Brakerski et al. [14] first constructed the BFV scheme that is based on Learning With Error (LWE) assumption [11], where the same modulus is utilized throughout the multiplication process. The main drawback of their work is high computational complexity due to LWE.

To its improvement, Fan et al. [12] port the Brakerski scheme into Ring Learning With Error (RLWE) and performs some optimization. First, the authors reduce the sample space size for secret key and thereafter during relinearization,⁵ Fan et al. [12] divide the key into smaller chunks and perform the homomorphic multiplication using modulus switching. The main limitation of [12]’s work is its inability to perform coefficient-wise division and rounding (when the coefficients of ciphertext polynomials are large) that is required during decryption and homomorphic multiplication.

Later on, Bajard et al. [15] proposed the Residue Number System (RNS) based setting where large modulus is divided into smaller coprime moduli and thereafter multiplications are performed. Halevi et al. [16] extends the work of Bajard

⁴ We believe our idea can be utilized for BGV scheme.

⁵ a way of converting degree two ciphertext into degree one ciphertext.

et al. [15] by manipulating large coefficients in the ciphertexts polynomials by considering Chinese Remainder Theorem (CRT).

Thereafter, Kim et al. [13] proposed two approaches to multiply the BFV ciphertexts. In the first approach, the authors proposed lazy scaling.⁶ In the second approach, the author proposed modulus switching of any single ciphertext with respect to another modulus that helps in reducing the total number of applied fast base conversions (FBCs).⁷

To optimize the noise growth, we improve the work of [12] and provide the theoretical homomorphic multiplication noise bounds using the work of Kim et al. [13].

1.3 Our Contribution

We propose an improved noise growth in BFV homomorphic encryption and present its application in ciphertext-ciphertext multiplication.

- First, we present an improved noise bound on Fan’s scheme by selecting secret key from χ_{err} (where χ_{err} is discrete Gaussian distribution) and error from R_2 (where the maximum norm of error is reduced from B_{err} to 1). Later, we empirically check the hardness of our proposed scheme using lattice estimator that achieves more than 128-bit security for desired attacks (as discussed in Table 2).
- Second, we revisit the impact of new noise bound on BFV homomorphic multiplication proposed by Kim et al. [13].

1.4 Our Organization

The preliminaries of our work are discussed in Section 2. In Section 3, we present the proposed methodology of our work. Later on, we show the security of our work in Section 4. In Section 5, we present the comparison with respect to various distribution settings. Section 6 discusses conclusion of our work.

2 Preliminaries

We first introduce the mathematical notations used in this work. Thereafter, Ring Learning With Errors (RLWE) assumption is explained (that provides security to the applied homomorphic encryption scheme). Thereafter, we discuss Fan et al.’s BFV homomorphic encryption scheme. Afterward, we discuss the fast base conversion that helps in converting one modulus space to another. Later on, we discuss Kim et al.’s BFV homomorphic multiplication.

⁶ Multiplication with the help of $\lfloor \frac{Q}{t} m \rfloor$ instead of $\lfloor \frac{Q}{t} \rfloor m$.

⁷ FBC allows to convert one modulus space to another [17].

2.1 Basic Notations

In this work, we utilize the polynomial ring $R = \mathbb{Z}[x]/\langle f(x) \rangle$ (where $f(x)$ is monic irreducible polynomial of degree n). Commonly used monic polynomial is $f(x) = x^n + 1$. Elements of ring R are denoted by lowercase bold letters (say $\mathbf{a}$). a_i represents coefficient of element $\mathbf{a} \in R$, that is, $\mathbf{a} = \sum_{i=0}^{n-1} a_i \cdot x^i$. We denote the l_∞ norm of $\mathbf{a}$ as $\max_i \{a_i\}$. There exist a constant δ_R (where δ_R is also known as ring expansion factor) such that $\|\mathbf{a} \cdot \mathbf{b}\|_\infty \leq \delta_R \|\mathbf{a}\|_\infty \|\mathbf{b}\|_\infty$, $\forall (\mathbf{a}, \mathbf{b}) \in R^2$. It is well known that for $R = \mathbb{Z}[x]/x^n + 1$, $\delta_R = n$. R_Q indicates ciphertext modulus where Q is an integer (where $R_Q = R/QR$). R_t signifies plaintext modulus (where t is an integer such that $Q > t$). $[\mathbf{a}]_Q$ indicates unique integer in R_Q such that $[\mathbf{a}]_Q = \mathbf{a} \bmod Q$. For $\mathbf{a} \in \mathbb{R}$, $\lfloor \mathbf{a} \rfloor$ indicates rounding to closet integer, $\lceil \mathbf{a} \rceil$ and $\lfloor \mathbf{a} \rfloor$ indicates rounding up and down.

Let χ be a probability distribution over R . We follow $\mathbf{a} \in \chi$ that represent $\mathbf{a} \in R$. Let χ_{key} denotes uniform ternary distribution where all the coefficients $\mathbf{a} \leftarrow \chi_{key}$ are selected uniformly from $\{-1, 0, 1\}$. χ_{err} indicates truncated discrete Gaussian distribution [18]. The distributions χ_{key} , χ_{err} are bounded by $[-B_{key}, B_{key}]$, $[-B_{err}, B_{err}]$ having standard deviation σ_{key} , σ_{err} respectively. Table 1 summarizes the significant notations used in this work.

Notation	Description
R	polynomial ring
n	degree of polynomial
$\mathbf{s}$	secret key
pk	public key
$\mathbf{m}$	message
$\mathbf{e}$	noise
$\mathbf{ct}$	ciphertext
χ_{key}	uniform ternary distribution bounded by $[-B_{key}, B_{key}]$
σ_{key}	standard deviation of distribution χ_{key}
χ_{err}	truncated discrete Gaussian distribution bounded by $[-B_{err}, B_{err}]$
σ_{err}	standard deviation of distribution χ_{err}
R_t	plaintext modulus
R_Q	ciphertext modulus
δ_R	ring expansion factor
$[\mathbf{a}]_Q$	$\mathbf{a} \bmod Q$

Table 1. Notations used in this work.

2.2 Ring Learning With Error (RLWE)

The security of homomorphic encryption is based on RLWE assumption. RLWE is a ring version of LWE [19]. Next we discuss the definition of decision-RLWE problem that is used in this work as follow.

Definition 1 (Decision-RLWE). *Let λ be the security parameter. Given a monic irreducible polynomial $f(x)$ of degree n , set R as a polynomial ring. For a randomly chosen element $\mathbf{s} \in R_Q$ and distribution $\chi = \chi(\lambda)$ (where χ be the distribution on R_Q), select a uniformly random element $\mathbf{a} \in R_Q$ and an error $\mathbf{e} \in \chi$ such that the following equation holds.*

$$\mathbf{b} = \mathbf{a} \cdot \mathbf{s} + \mathbf{e} \pmod{Q} \quad (1)$$

Remark 1. It is cryptographically hard to guess the secret key $\mathbf{s}$ from above consequences [11].

2.3 Overview of BFV Scheme (Fan et al. [12])

Let $\mathbf{m}$ be the message such that $\mathbf{m} \in R_t$. $\lambda \in R_t$ is the security parameter. The encryption scheme is performed in various steps as follows.

- SecretKeyGen(1^λ): sample $\mathbf{s} \in R_2$.⁸
- PublicKeyGen($\mathbf{s}$): sample $\mathbf{a} \in R_Q$, $\mathbf{e} \in \chi_{err}$ and output

$$pk = ([-(\mathbf{a} \cdot \mathbf{s} + \mathbf{e})]_Q, \mathbf{a}).$$

- Encrypt($pk, \mathbf{m}$): let $\mathbf{p}_0 = pk[0]$, $\mathbf{p}_1 = pk[1]$, sample $\mathbf{u} \in R_2$, $\mathbf{e}_1, \mathbf{e}_2 \in \chi_{err}$ and return

$$ct = ([\mathbf{p}_0 \cdot \mathbf{u} + \mathbf{e}_1 + \Delta \cdot \mathbf{m}]_Q, [\mathbf{p}_1 \cdot \mathbf{u} + \mathbf{e}_2]_Q)$$

(where $\Delta = \lfloor \frac{Q}{t} \rfloor$).

- Decrypt($\mathbf{s}, ct$): let $\mathbf{c}_0 = ct[0]$, $\mathbf{c}_1 = ct[1]$ and compute

$$\left[\left\lfloor \frac{t \cdot [\mathbf{c}_0 + \mathbf{c}_1 \cdot \mathbf{s}]_Q}{Q} \right\rfloor \right]_t$$

Lemma 1. *Using above encryption scheme and assuming $\|\chi_{err}\|_\infty < B_{err}$ and $\|\mathbf{s}\|_\infty = \|\mathbf{u}\|_\infty \leq 1$, we have that*

$$ct(\mathbf{s}) = \mathbf{c}_0 + \mathbf{c}_1 \cdot \mathbf{s} = \Delta \mathbf{m} + \mathbf{v} \pmod{Q}, \quad (2)$$

with $\mathbf{v} = \mathbf{e} \cdot \mathbf{u} + \mathbf{e}_1 + \mathbf{e}_2 \cdot \mathbf{s}$. This implies that for given bound in Eq. 3, decryption works correctly.

$$\|\mathbf{v}\|_\infty \leq 2\delta_R B_{err} + B_{err} \quad (3)$$

Remark 2. For correct decryption $\|\mathbf{v}\|_\infty \leq \frac{Q}{2t}$.

⁸ In standard LWE one can take $\mathbf{s}$ from any distribution, as long as distribution has sufficient entropy [20].

2.4 Fast Base Conversion (FBC)

At some point, to control the noise growth, a polynomial to be converted from R_Q to $R_{\mathcal{B}}$.⁹ To achieve such kind of operations efficiently (with respect to time), FBC is widely used (as suggested by [15]). The following equation convert residues of $a \in [0, Q)$ from modulus Q to modulus $\mathcal{B}$ (where Q is coprime to $\mathcal{B}$).

$$FBC(a, Q, \mathcal{B}) = \sum_{i=1}^k \left[a \left(\frac{Q}{q_i} \right)^{-1} \right]_{q_i} \frac{Q}{q_i} \bmod \mathcal{B}_j \quad (4)$$

(where $Q = q_1 \cdots q_k$, $\mathcal{B} = \mathcal{B}_1 \cdots \mathcal{B}_{k'}$). Note that Eq. (4) yields $[a]_Q + KQ \bmod \mathcal{B}$ (where $K \in [0, k-1]$). When this result is divided by Q , the error caused by this Q -overflow K can be neglected most of the time. This overflow error can be reduced with the γ -correction technique of [15], or floating point instructions technique [16].

2.5 Overview of Ciphertext Multiplication (Kim et al. [13])

In this section, we present the ciphertext multiplication algorithm from Section 3.2 of the work by Kim et al. [13], which represents the most recent approach to multiplication in the BFV scheme.

Consider two ciphertexts as follows.

$$ct_1(s) = \frac{Q}{t} [\mathbf{m}_1]_t + \mathbf{v}_1 \bmod Q, \quad (5)$$

and

$$ct_2(s)' = \frac{Q}{t} [\mathbf{m}_2]_t + \mathbf{v}_2 \bmod Q. \quad (6)$$

Thereafter, instead of multiplying two ciphertext under mod Q and dealing with a multiple of Q^2 modulo $Q\mathcal{B}$, Kim et al. [13] switch any one of the ciphertext into another modulus $\mathcal{B}$ (where $\mathcal{B} = \mathcal{B}_1 \cdots \mathcal{B}_{k'}$). This notion obtain a multiple of $Q\mathcal{B}$ during multiplication that helps to remove some of the part of noise (meaning $Q\mathcal{B}$ modulo $Q\mathcal{B}$), and also reduces the number of FBCs. Thus, modulus switching is performed over $ct_2(s)'$ with respect to modulus $\mathcal{B}$ (here $\mathcal{B} \approx Q$) as follows.

$$\begin{aligned} ct_2(s) &= \left\lfloor \frac{\mathcal{B}}{Q} ct_2(s)' \right\rfloor \bmod \mathcal{B} \\ &= \frac{\mathcal{B}}{t} [\mathbf{m}_2]_t + \frac{\mathcal{B}\mathbf{v}_2}{Q} + \epsilon_r \bmod \mathcal{B} \end{aligned} \quad (7)$$

(where $\epsilon_r = ct_2(s) - \mathcal{B}ct_2(s)'/Q$ has its norm bounded by $(1 + \delta_R B_{key})/2$). Afterward, FBCs are applied to convert one modulus to another to perform computation in $\mathcal{B}Q$ modulus as follows.

$$ct_1(s) = \frac{Q}{t} [\mathbf{m}_1]_t + \mathbf{v}_1 + K_1 Q \bmod \mathcal{B}Q, \quad (8)$$

⁹ An approach to delay bootstrapping or regulate the number of homomorphic operations by modifying the sample space size, either by increasing or decreasing it.

and

$$ct_2(s) = \frac{\mathcal{B}}{t}[\mathbf{m}_2]_t + \frac{\mathcal{B}}{Q}\mathbf{v}_2 + K_2\mathcal{B} + \epsilon_r \bmod \mathcal{B}Q. \quad (9)$$

Thereafter, Eq. (8) is multiplied with Eq. (9) as follows.

$$\begin{aligned} ct_{mult}(s) &= \frac{\mathcal{B}Q}{t^2}[\mathbf{m}_1\mathbf{m}_2]_t + \frac{\mathcal{B}}{t}\tilde{\mathbf{v}} + \frac{\mathcal{B}Q}{t}\tilde{K} \\ &\quad + \epsilon_r \left(\frac{Q}{t}[\mathbf{m}_1]_t + \mathbf{v}_1 + K_1Q \right) \bmod \mathcal{B}Q, \end{aligned} \quad (10)$$

where

$$\begin{aligned} \tilde{\mathbf{v}} &= [\mathbf{m}_1]_t \cdot \mathbf{v}_1 + [\mathbf{m}_2]_t \cdot \mathbf{v}_2 + \frac{t}{Q}\mathbf{v}_1\mathbf{v}_2 + t(\mathbf{v}_1K_2 + \mathbf{v}_2K_1), \\ \tilde{K} &= \mathbf{m}_1K_2 + \mathbf{m}_2K_1 + tK_1K_2 + r_m. \end{aligned}$$

Here, $\|r_m\|_\infty \leq \delta_R t/2$, and note that $K_i = (ct_i(s) - (Q/t)[\mathbf{m}_i]_t - \mathbf{v}_i)/Q$ (where $1 \leq i \leq 2$) having norm bounded by $\|K\|_\infty \leq (\delta_R B_{key} + 3)/2$. Note that to get back the valid ciphertext under modulo Q , one must scale down the result by $t/\mathcal{B}$ instead of t/Q as follows.

$$\begin{aligned} ct_{scale}(s) &= \frac{t}{\mathcal{B}}ct_{mult}(s) = \frac{Q}{t}[\mathbf{m}_1\mathbf{m}_2]_t + \tilde{\mathbf{v}} + Q\tilde{K} \\ &\quad + \frac{t\epsilon_r}{\mathcal{B}} \cdot \left(\frac{Q}{t}[\mathbf{m}_1]_t + \mathbf{v}_1 + K_1Q \right) \end{aligned} \quad (11)$$

We remark that 1) the multiple of Q will vanish modulo Q but the above Eq. (11) produces the scaling error (say $\mathbf{v}_r$), where the norm is bounded by $\|\mathbf{v}_r\|_\infty = (1 + \delta_R B_{key} + \delta_R^2 B_{key}^2)/2$, and 2) the overflow error (that comes due to FBCs) is insignificant as suggested by [16] that can be corrected with the help of γ -correction technique [15], or floating point instructions [16]. Therefore, the noise in this scheme is bounded by following equation.

$$\begin{aligned} \|\mathbf{v}_{mult}\|_\infty &\leq \frac{\delta_R t}{2} \left(\frac{(2\delta_R B_{err} B_{key} + B_{err})^2}{Q} + (4 + \delta_R B_{key})(4\delta_R B_{err} B_{key} + 2B_{err}) \right) \\ &\quad + \frac{1 + \delta_R B_{key} + \delta_R^2 B_{key}^2}{2} \\ &\quad + \frac{t\delta_R(\delta_R B_{key} + 1)}{2\mathcal{B}} \left((2\delta_R B_{err} B_{key} + B_{err}) + \frac{Q(\delta_R B_{key} + 4)}{2} \right) \end{aligned} \quad (12)$$

Where

$$\begin{aligned} \mathbf{v}_i &= \mathbf{e} \cdot \mathbf{u} + \mathbf{e}_1 + \mathbf{e}_2 \cdot \mathbf{s}, \\ \mathbf{e}, \mathbf{e}_1, \text{ and } \mathbf{e}_2 &\in \chi_{err}, \mathbf{u}, \text{ and } \mathbf{s} \in \chi_{key}. \end{aligned}$$

Thus,

$$\|\mathbf{v}_i\|_\infty \leq 2\delta_R B_{err} B_{key} + B_{err}.$$

(here $\|\mathbf{v}_1\|_\infty = \|\mathbf{v}_2\|_\infty \leq 2\delta_R B_{err} B_{key} + B_{err}$).

3 Proposed Methodology

In this section, first we propose our improvement by swapping the distribution space and thereafter we discuss the impact of noise growth in Kim et al.'s multiplication.

3.1 Our Improvement

Instead of sampling secret key $\mathbf{s}, \mathbf{u} \in R_2$, we sample errors $\mathbf{e}, \mathbf{e}_1$, and $\mathbf{e}_2 \in R_2$ (such that the l_∞ norm of R_2 is 1). We note that secret key and $\mathbf{u}$ sampled from χ_{err} . The security implications of this improvement seem to be minor: 1) in standard LWE one can take secret key from any distribution having sufficient entropy [20], and 2) the LWE assumption remains hard even when the errors are small, that is, linear in the dimension of the LWE secret [21]. Thus, the obtained noise bound on BFV encryption is as follows.

$$\|\mathbf{v}\|_\infty \leq 2\delta_R B_{err} + 1, \quad (13)$$

where

$$\begin{aligned} \mathbf{v} &= \mathbf{e} \cdot \mathbf{u} + \mathbf{e}_1 + \mathbf{e}_2 \cdot \mathbf{s}, \\ \|\mathbf{e}\|_\infty &= \|\mathbf{e}_1\|_\infty = \|\mathbf{e}_2\|_\infty \leq 1, \\ \|\mathbf{s}\|_\infty &= \|\mathbf{u}\|_\infty \leq B_{err}. \end{aligned}$$

Hence, the noise bound present in Eq. (13) is less as compared to obtained noise bound in Eq. (3).

3.2 Noise Growth on Kim et al.'s Multiplication

In this section, we perform the BFV homomorphic multiplication using the new noise bound obtained in Section 3.1. We first recall Eq. (11) as follows.

$$ct(s)_{scale} = \frac{Q}{t} [\mathbf{m}_1 \mathbf{m}_2]_t + \tilde{\mathbf{v}} + Q\tilde{K} + \frac{t\epsilon_r}{B} \cdot \left(\frac{Q}{t} [\mathbf{m}_1]_t + \mathbf{v}_1 + K_1 Q \right)$$

Thus, the achieved noise growth is as follows.

$$\begin{aligned} \|\mathbf{v}_{mult}\|_\infty &\leq \frac{\delta_R t}{2} \left(\frac{(2\delta_R B_{err} + 1)^2}{Q} + (4 + \delta_R B_{err})(4\delta_R B_{err} + 2) \right) \\ &\quad + \frac{1 + \delta_R B_{err} + \delta_R^2 B_{err}^2}{2} + \frac{t\delta_R(\delta_R B_{err} + 1)}{2B} \left((2\delta_R B_{err} + 1) + \frac{Q(\delta_R B_{err} + 4)}{2} \right) \end{aligned} \quad (14)$$

where

$$\begin{aligned} \|K_i\|_\infty &\leq (\delta_R B_{err} + 3)/2, \\ \|\mathbf{v}_i\|_\infty &\leq 2\delta_R B_{err} + 1, \\ \|\mathbf{v}_r\|_\infty &\leq (1 + \delta_R B_{err} + \delta_R^2 B_{err}^2)/2, \end{aligned}$$

and

$$\|\epsilon_R\|_\infty \leq (1 + \delta_R B_{err})/2.$$

4 Security

The security of the proposed work is computed in two steps. In the first step, we discuss the hardness of RLWE assumption. Thereafter, the security of proposed work is verified using lattice estimator.

4.1 RLWE base Encryption

The hardness of the Ring Learning With Errors (RLWE) problem is based on the difficulty of solving the Learning With Errors (LWE) problem in structured algebraic settings, specifically in polynomial rings. The problem remains hard under worst-case reductions from ideal lattice problems, making it a strong foundation for post-quantum cryptography [11]. The security of RLWE relies on the inability to distinguish noisy ring-based linear equations from uniform randomness. LWE is also hard even with small parameters, as it can be reduced to the Short Integer Solution (SIS) problem (for more details, we refer the readers to [21]).

4.2 Security Analysis using Lattice Estimator

In this section, we perform our task on Intel(R) Core(TM) i5-10500 CPU @ 3.10GHz 3.10 GHz. The installed memory is 16 GB.

We run our verification on Kyber 512.¹⁰ The dimension of LWE problem is considered as $2 * 256$, whereas number of samples are also considered as $2 * 256$, secret key is selected from discrete Gaussian distribution having standard deviation 3.19 (as suggested in [Section 2.1.5][22]) and error is selected from R_2 (where R_2 is considered binary distribution having coefficients that helps in reducing the l_∞ norm, that is, 1). Table 2 represent running operations required against various attacks [23,24]. From Table 2, the desired attacks required more

Attack	Running operation(s)
bkw	$2^{143.1}$
usvp	$2^{142.4}$
bdd	$2^{139.5}$
dual	$2^{149.3}$
dual_hybrid	$2^{133.5}$

Table 2. Here bkw represent Blum-Kalai-Wasserman attack [25]. USVP represent Unique Shortest Vector Problem attack [26]. BDD represent Bounded Distance Decoding attack [27]. dual represent dual attack [28], and dual_hybrid represent Dual Hybrid attack [29].

than 128-bit security. Thus, under desired conditions the proposed scheme is secure against all the desired attacks on LWE assumption.

¹⁰ It is a parameter set for kyber, a lattice based key encapsulation mechanism (KEM) designed for post quantum cryptography that works over 128-bit security.

5 Comparison With Respect to Various Parameter Settings

In this section, we run our verification on Kyber 512 against desired security attacks [23,24]. We first compared, Fan et al.'s, and Kim et al.'s setting with different relationship between number of samples (say N) and dimension of LWE secret (say M) (as shown in Table [3, 4, 5]). Thereafter, we compare our setting with some other distribution with respect to secret key, error distribution (as shown in Table 6). We remark that, the standard deviation of discrete Gaussian distribution is 3.19 [22].

Attack	Running Operation(s) (Our Setting)		
	$s \in \chi_{err}, \quad e \in R_2$		
	$N = 2 \times 256$ $M = 2 \times 256$	$N = 2 \times 256$ $M = 1 \times 256$	$N = 2 \times 256$ $M = 4 \times 256$
bkw	$2^{143.1}$	$2^{143.1}$	Failed
usvp	$2^{142.4}$	$2^{142.4}$	$2^{117.5}$
bdd	$2^{139.5}$	$2^{139.5}$	$2^{114.5}$
dual	$2^{149.3}$	$2^{149.3}$	$2^{121.9}$
dual_hybrid	$2^{133.5}$	$2^{133.5}$	$2^{111.1}$

Table 3. Here, $\mathbf{s}$ and $\mathbf{e}$ represent the secret key and error, where χ_{err} represents the discrete Gaussian distribution and R_2 represents binary distribution. The attacks include: Blum-Kalai-Wasserman (bkw) [25], Unique Shortest Vector Problem (usvp) [26], Bounded Distance Decoding (bdd) [27], dual attack (dual) [28], and Dual Hybrid attack (dual_hybrid) [29].

Attack	Running Operation(s) (Fan et al. [12] setting)		
	$s \in R_2, \quad e \in \chi_{err}$		
	$N = 2 \times 256$ $M = 2 \times 256$	$N = 2 \times 256$ $M = 1 \times 256$	$N = 2 \times 256$ $M = 4 \times 256$
bkw	$2^{143.1}$	$2^{143.1}$	$2^{143.1}$
usvp	$2^{142.4}$	Failed	$2^{142.4}$
bdd	$2^{139.5}$	Failed	$2^{139.5}$
dual	$2^{149.3}$	$2^{133.5}$	$2^{149.3}$
dual_hybrid	$2^{133.5}$	$2^{133.5}$	$2^{133.5}$

Table 4. Here, $\mathbf{s}$ and $\mathbf{e}$ represent the secret key and error, where χ_{err} represents the discrete Gaussian distribution and R_2 represent binary distribution. The attacks include: Blum-Kalai-Wasserman (bkw) [25], Unique Shortest Vector Problem (usvp) [26], Bounded Distance Decoding (bdd) [27], dual attack (dual) [28], and Dual Hybrid attack (dual_hybrid) [29].

Attack	Running Operation(s) (Kim et al. [13] setting)		
	$s \in R_3, e \in \chi_{err}$		
	$N = 2 \times 256$ $M = 2 \times 256$	$N = 2 \times 256$ $M = 1 \times 256$	$N = 2 \times 256$ $M = 4 \times 256$
bkw	$2^{166.7}$	$2^{161.1}$	$2^{166.7}$
usvp	$2^{153.2}$	Failed	$2^{153.2}$
bdd	$2^{149.8}$	Failed	$2^{149.5}$
dual	$2^{160.4}$	$2^{155.9}$	$2^{160.4}$
dual_hybrid	$2^{145.7}$	$2^{145.7}$	$2^{133.5}$

Table 5. Here, s and e represent the secret key and error, where χ_{err} represents the discrete Gaussian distribution and R_3 represents uniform ternary distribution. The attacks include: Blum-Kalai-Wasserman (bkw) [25], Unique Shortest Vector Problem (usvp) [26], Bounded Distance Decoding (bdd) [27], dual attack (dual) [28], and Dual Hybrid attack (dual_hybrid) [29].

Attack	Running Operation(s)						
	$N = 2 \times 256, M = 2 \times 256$						
	$s \in \chi_{err}$ $e \in R_2$	$s \in R_2$ $e \in \chi_{err}$	$s \in R_3$ $e \in \chi_{err}$	$s \in \chi_{err}$ $e \in R_3$	$s \in R_2$ $e \in R_2$	$s \in R_3$ $e \in R_3$	$s \in \chi_{err}$ $e \in \chi_{err}$
bkw	$2^{143.1}$	$2^{143.1}$	$2^{166.7}$	$2^{166.7}$	Failed	$2^{132.2}$	$2^{202.8}$
usvp	$2^{142.4}$	$2^{142.2}$	$2^{153.2}$	$2^{153.2}$	$2^{117.5}$	$2^{123.3}$	$2^{185.2}$
bdd	$2^{139.5}$	$2^{139.5}$	$2^{149.8}$	$2^{149.8}$	$2^{114.5}$	$2^{120.2}$	$2^{180.6}$
dual	$2^{149.3}$	$2^{149.3}$	$2^{160.4}$	$2^{160.4}$	$2^{121.9}$	$2^{128.1}$	$2^{195.3}$
dual_hybrid	$2^{133.5}$	$2^{133.5}$	$2^{145.7}$	$2^{145.7}$	$2^{111.1}$	$2^{116.2}$	$2^{259.7}$

Table 6. Comparison of running operations for different attack methods under various distributions of secret key and error. Here, s and e represent the secret key and error, where χ_{err} represents the discrete Gaussian distribution, R_2 represents binary distribution, and R_3 represents uniform ternary distribution. The attacks include: Blum-Kalai-Wasserman (bkw) [25], Unique Shortest Vector Problem (usvp) [26], Bounded Distance Decoding (bdd) [27], dual attack (dual) [28], and Dual Hybrid attack (dual_hybrid) [29].

From the above consequences, Table 3 indicates that when $s \in \chi_{err}$, $e \in R_2$, and $N = M$, $N > M$ the proposed scheme achieves more than 128-bit security. However, in Table 4, under Fan et al.'s setting ($s \in R_2$, $e \in \chi_{err}$), the scheme is secure whenever $N = M$ and $N < M$. In Table 5, under Kim et al.'s setting ($s \in R_3$, $e \in \chi_{err}$), the scheme is secured whenever $N = M$ and $N < M$. Table 6 demonstrates that the RLWE scheme is not secure whenever both secret key and error are sampled from either binary (R_2) or uniform ternary distribution (R_3).

6 Conclusion

In this paper, we demonstrate an improvement over existing BFV homomorphic encryption schemes in terms of noise growth. We improve the noise bound in Fan’s scheme by swapping the distribution space for secret key and error. Later, we show the BFV homomorphic multiplication using Kim’s approach and computes the new noise bound. The resulting noise bound is less as compared to other existing works, which helps delay bootstrapping—an operation considered expensive in homomorphic encryption.

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